In [1]:

```
import numpy as np
```

In [2]:

```
import pandas as pd
```

In [3]:

```
import matplotlib.pyplot as plt
```

In [4]:

```
import seaborn as sns
```

Load data

In [5]:

```
df=pd.read_csv(r'C:\Users\M A H N A Z\Downloads\ipl_2026_deliveries.csv')
```

In [6]:

```
df
```

Out[6]:

	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
0	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
1	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
2	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
3	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
4	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
...	...	...	...	...	...	...	...	...	...
3512	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3513	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3514	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3515	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

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	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
3516	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

3517 rows × 21 columns

In [7]:

```
df.shape #rows & columns
```

Out[7]:

```
(3517, 21)
```

In [8]:

```
df.columns
```

Out[8]:

```
Index(['match_id', 'season', 'phase', 'match_no', 'date', 'venue',
       'batting_team', 'bowling_team', 'innings', 'over', 'striker', 'bowler',
       'runs_of_bat', 'extras', 'wide', 'legbyes', 'byes', 'noballs',
       'wicket_type', 'player_dismissed', 'fielder'],
      dtype='str')
```

In [9]:

```
df.info() #to get basic information about data
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3517 entries, 0 to 3516
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   match_id          3517 non-null   int64
 1   season            3517 non-null   int64
 2   phase             3517 non-null   str
 3   match_no          3517 non-null   int64
 4   date              3517 non-null   str
 5   venue             3517 non-null   str
 6   batting_team      3517 non-null   str
 7   bowling_team      3517 non-null   str
 8   innings           3517 non-null   int64
 9   over              3517 non-null   float64
 10  striker           3517 non-null   str
 11  bowler            3517 non-null   str
 12  runs_of_bat       3517 non-null   int64
 13  extras            3517 non-null   int64
 14  wide              3517 non-null   int64
 15  legbyes           3517 non-null   int64
 16  byes              3517 non-null   int64
 17  noballs           3517 non-null   int64
 18  wicket_type       192 non-null    str
 19  player_dismissed  188 non-null    str
 20  fielder           152 non-null    str
dtypes: float64(1), int64(10), str(10)
memory usage: 577.1 KB
```

In [10]:

```
#to check missing values
df.isnull().sum()
```

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Out[10]:

```
match_id               0
season                 0
phase                  0
match_no               0
date                   0
venue                  0
batting_team           0
bowling_team           0
innings                0
over                   0
striker                0
bowler                 0
runs_of_bat            0
extras                 0
wide                   0
legbyes                0
byes                   0
noballs                0
wicket_type         3325
player_dismissed    3329
fielder             3365
dtype: int64
```

How many matches are played?

In [11]:

```
df['match_id'].unique()
```

Out[11]:

```
array([202601, 202602, 202603, 202604, 202605, 202606, 202607, 202608,
       202609, 202610, 202611, 202612, 202613, 202614, 202615, 202616])
```

What and how many teams are playing?

In [12]:

```
df['batting_team'].unique()
```

Out[12]:

```
<StringArray>
['SRH', 'RCB', 'KKR', 'MI', 'CSK', 'RR', 'GT', 'PBKS', 'LSG', 'DC']
Length: 10, dtype: str
```

In [13]:

```
df['batting_team'].nunique()
```

Out[13]:

```
10
```

Feature engineering- creating new columns

In [14]:

```
df['Total_runs']=df['runs_of_bat']+df['extras'] #total run on each ball
df.head()
```

Out[14]:

	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings	o
0	202601	2026	Group Stage	1	Mar 28,	M.Chinnaswamy Stadium,	SRH	RCB	1	

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	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings	o
					2026	Bengaluru				
1	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
2	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
3	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
4	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	

5 rows × 22 columns

In [15]:

```
df['is_six']=0
df['is_four']=0
df['is_wicket']=0
df.loc[df['runs_of_bat']==6, 'is_six']=1
df.loc[df['runs_of_bat']==4, 'is_four']=1
df.loc[df['wicket_type'].notnull(), 'is_wicket']=1
df.head(20)
```

Out[15]:

	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings	
0	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
1	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
2	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
3	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
4	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
5	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
6	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
7	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	

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	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
8	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
9	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
10	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
11	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
12	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
13	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
14	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
15	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
16	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
17	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
18	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
19	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1

20 rows × 25 columns

In [16]:

```
df['over']=df['over']+1
df.head()
```

Out[16]:

	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
0	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
1	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1

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	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings	o
2	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
3	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	
4	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1	

5 rows × 25 columns

In [17]:

```
def get_phase(over):
    if over<=6:
        return 'Powerplay(1-6)'
    elif over<=15:
        return 'Middle(7-15)'
    else:
        return 'Death(16-20)'
df['phase_group']=df['over'].apply(get_phase)
df
```

Out[17]:

	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
0	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
1	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
2	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
3	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
4	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
...	...	...	...	...	...	...	...	...	...
3512	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3513	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3514	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

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	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
3515	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3516	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

3517 rows × 26 columns

Batting Analysis

Which teams scored most runs?

In [18]:

```
df
```

Out[18]:

	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
0	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
1	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
2	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
3	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
4	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
...	...	...	...	...	...	...	...	...	...
3512	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3513	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3514	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3515	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3516	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

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3517 rows × 26 columns

In [19]:

```
df1=df.groupby('batting_team')['Total_runs'].sum().sort_values(ascending=False).reset_in
```

In [20]:

```
df1
```

Out[20]:

	batting_team	Total_runs
0	RR	690
1	RCB	654
2	KKR	586
3	SRH	585
4	GT	576
5	CSK	544
6	DC	516
7	MI	509
8	LSG	483
9	PBKS	374

In [21]:

```
plt.figure(figsize=(10,6))
plt.bar(df1['batting_team'],df1['Total_runs'], color="skyblue",edgecolor='black')
plt.title("Team wise total score")
plt.xlabel("Batting Team")
plt.ylabel("Total Score")
for x,y in zip(df1['batting_team'],df1['Total_runs']):
    plt.text(x=x,y=y,s=y,va='bottom')
plt.show()
```

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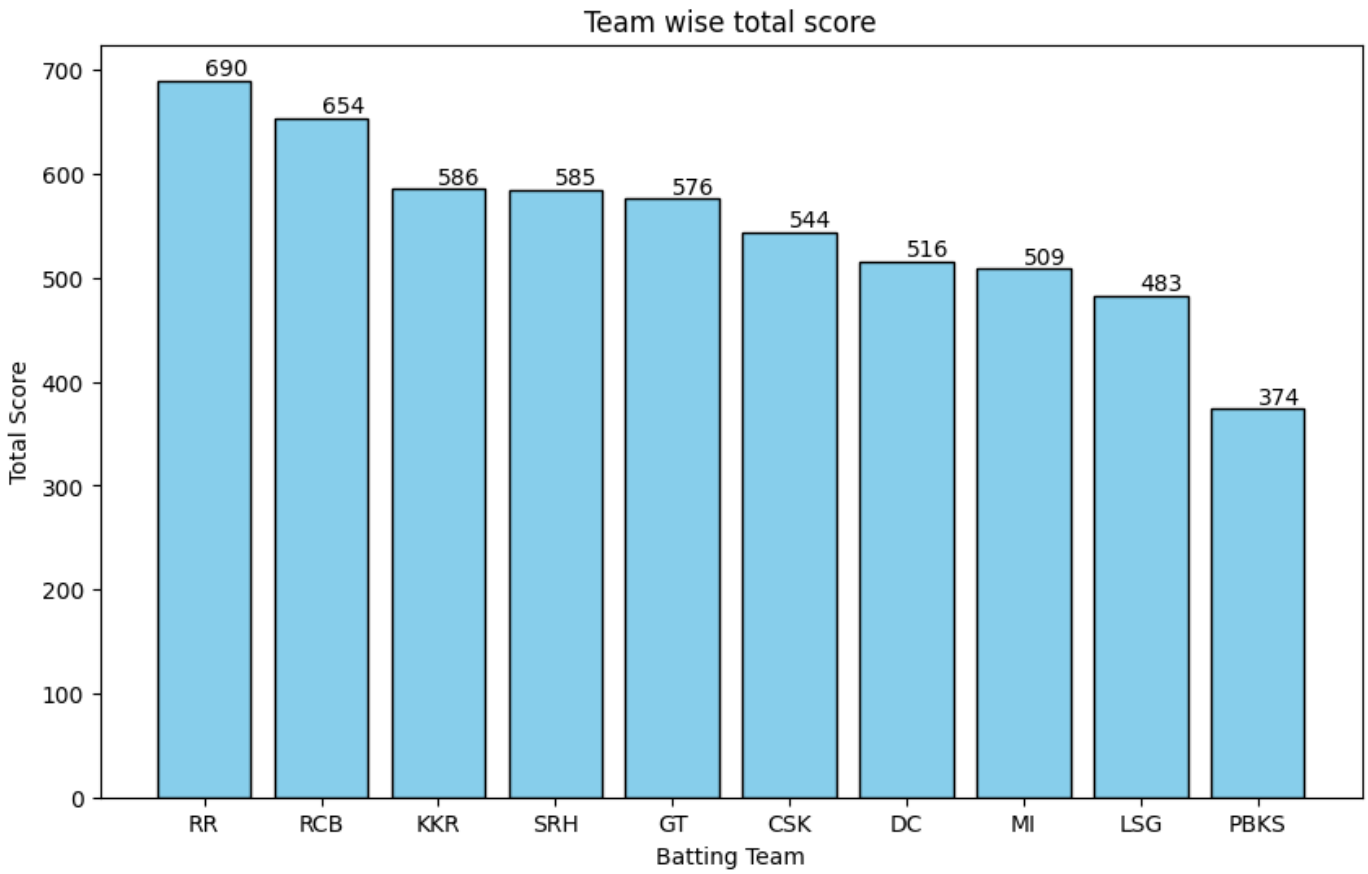

Teams with most sixs and fours

In [22]:

```
df
```

Out[22]:

	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
0	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
1	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
2	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
3	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
4	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
...	...	...	...	...	...	...	...	...	...
3512	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

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	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
3513	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3514	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3515	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3516	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

3517 rows × 26 columns

In [23]:

```
df1 = df.groupby('batting_team').agg(
    nos=('is_six', 'sum'),
    nof=('is_four', 'sum')
).reset_index()
df1['score of boundaries']=df1['nos']+df1['nof']
df1
```

Out[23]:

	batting_team	nos	nof	score of boundaries
0	CSK	23	50	73
1	DC	22	50	72
2	GT	23	55	78
3	KKR	24	57	81
4	LSG	19	44	63
5	MI	23	50	73
6	PBKS	23	29	52
7	RCB	39	51	90
8	RR	42	59	101
9	SRH	29	50	79

In [24]:

```
plt.figure(figsize=(15,6))

xval = np.arange(1, 11)

plt.bar(xval, df1['nos'], width=0.3, color='green', label='sixes', edgecolor='black')
plt.bar(xval+0.2, df1['nof'], width=0.3, color='lightgreen', label='fours', edgecolor='b
plt.bar(xval+0.4, df1['score of boundaries'], width=0.3, color='blue', label='total', ed

plt.legend()

# Add labels
```

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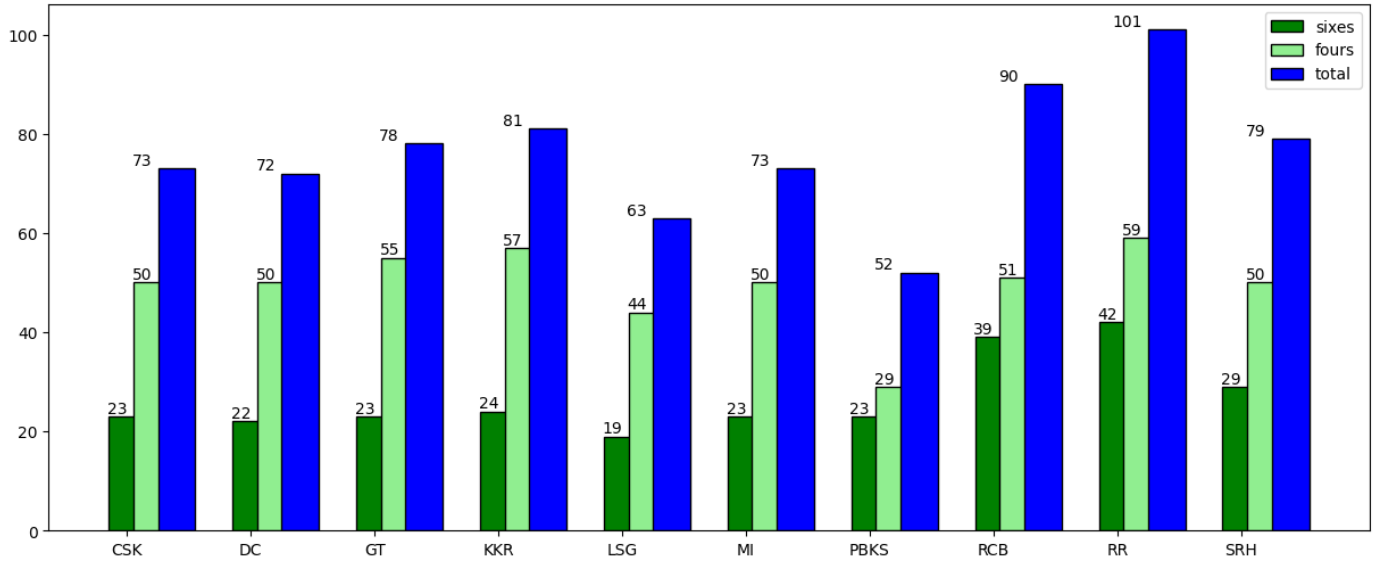
```
for x, y in zip(xval, df1['nos']):
    plt.text(x=x, y=y, s=y, va='bottom', ha='right')

for x, y in zip(xval+0.2, df1['nof']):
    plt.text(x=x, y=y, s=y, va='bottom', ha='right')
for x, y in zip(xval+0.2, df1['score of boundaries']):
    plt.text(x=x, y=y, s=y, va='bottom', ha='right')
plt.xticks(xval,df1['batting_team'])

plt.show()
```



How many runs did each team score in Powerplay , Middle & Death overs?

In [25]:

```
df
```

Out[25]:

	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
0	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
1	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
2	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
3	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
4	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
...	...	...	...	...	...	...	...	...	...
3512	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

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	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
3513	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3514	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3515	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3516	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

3517 rows × 26 columns

In [26]:

```
df1=df.groupby(['batting_team','phase_group']).agg(
    total=('Total_runs','sum')).reset_index()
df1=df1.sort_values(['batting_team','total'],ascending=False)
df1
```

Out[26]:

	batting_team	phase_group	total
28	SRH	Middle(7-15)	264
27	SRH	Death(16-20)	201
29	SRH	Powerplay(1-6)	120
25	RR	Middle(7-15)	324
26	RR	Powerplay(1-6)	265
24	RR	Death(16-20)	101
22	RCB	Middle(7-15)	278
21	RCB	Death(16-20)	208
23	RCB	Powerplay(1-6)	168
19	PBKS	Middle(7-15)	153
20	PBKS	Powerplay(1-6)	112
18	PBKS	Death(16-20)	109
16	MI	Middle(7-15)	256
17	MI	Powerplay(1-6)	142
15	MI	Death(16-20)	111
13	LSG	Middle(7-15)	211
12	LSG	Death(16-20)	148
14	LSG	Powerplay(1-6)	124
10	KKR	Middle(7-15)	238

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	batting_team	phase_group	total
11	KKR	Powerplay(1-6)	200
9	KKR	Death(16-20)	148
7	GT	Middle(7-15)	261
6	GT	Death(16-20)	160
8	GT	Powerplay(1-6)	155
4	DC	Middle(7-15)	252
3	DC	Death(16-20)	156
5	DC	Powerplay(1-6)	108
1	CSK	Middle(7-15)	224
0	CSK	Death(16-20)	171
2	CSK	Powerplay(1-6)	149

In [27]:

```
df1 = df.pivot_table(index='batting_team', columns='phase_group',values='Total_runs', ag
df1
```

Out[27]:

phase_group	batting_team	Death(16-20)	Middle(7-15)	Powerplay(1-6)
0	CSK	171	224	149
1	DC	156	252	108
2	GT	160	261	155
3	KKR	148	238	200
4	LSG	148	211	124
5	MI	111	256	142
6	PBKS	109	153	112
7	RCB	208	278	168
8	RR	101	324	265
9	SRH	201	264	120

In [28]:

```
plt.figure(figsize=(15,6))


# Dynamic x values
xval = np.arange(len(df1))

# Bars
plt.bar(xval, df1['Powerplay(1-6)'], width=0.3, color='green', label='Powerplay', edgeco
plt.bar(xval+0.2, df1['Middle(7-15)'], width=0.3, color='lightgreen', label='Middle', ed
plt.bar(xval+0.4, df1['Death(16-20)'], width=0.3, color='blue', label='Death', edgecolor

plt.legend()
```

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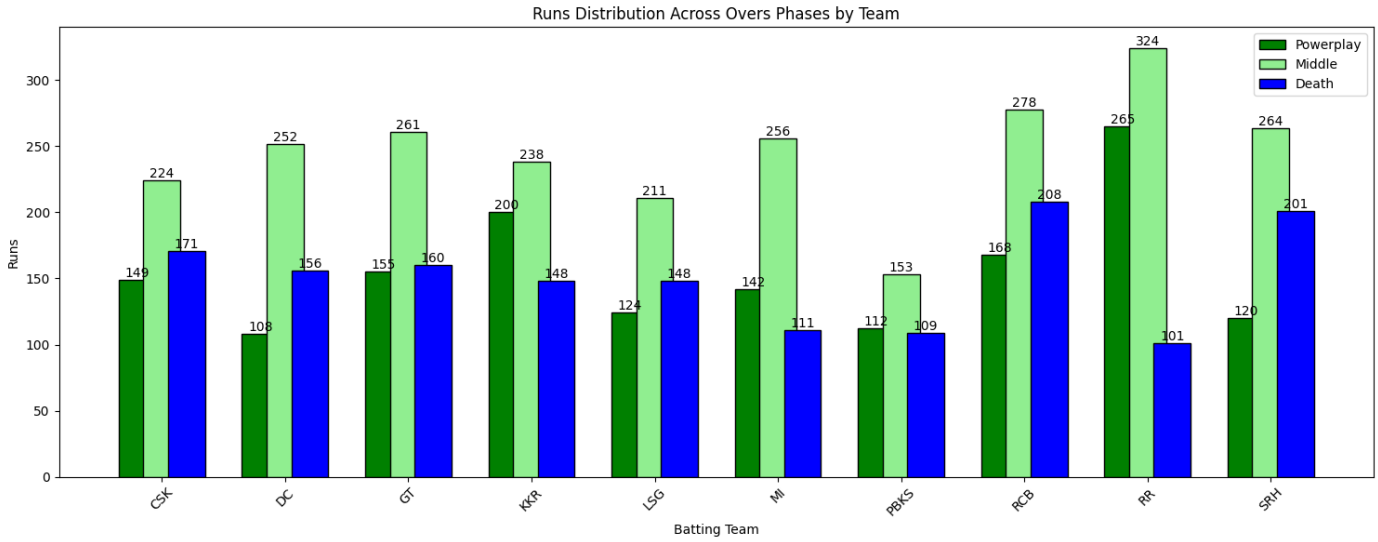
```
# Labels
for x, y in zip(xval, df1['Powerplay(1-6)']):
    plt.text(x=x, y=y, s=y, va='bottom', ha='center')

for x, y in zip(xval+0.2, df1['Middle(7-15)']):
    plt.text(x=x, y=y, s=y, va='bottom', ha='center')

for x, y in zip(xval+0.4, df1['Death(16-20)']):
    plt.text(x=x, y=y, s=y, va='bottom', ha='center')

# X-axis labels
plt.xticks(xval+0.2, df1['batting_team'], rotation=45)  # centered under grouped bars

plt.title("Runs Distribution Across Overs Phases by Team")
plt.xlabel("Batting Team")
plt.ylabel("Runs")

plt.tight_layout()
plt.show()
```



BatsMan Analysis

TOP 10 Scores in IPL2026

In [29]:

```
df
```

Out[29]:

	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
0	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
1	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
2	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1

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	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
3	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
4	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
...	...	...	...	...	...	...	...	...	...
3512	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3513	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3514	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3515	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3516	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

3517 rows × 26 columns

In [30]:

```
df1=df.groupby('striker').agg(
    total=('Total_runs','sum')
).reset_index()
df1=df1.sort_values('total',ascending=False).head(10)
df1
```

Out[30]:

	striker	total
101	Vaibhav Sooryavanshi	204
108	Yashasvi Jaiswal	193
22	Dhruv Jurel	185
80	Sameer Rizvi	175
4	Angkrish Raghuvanshi	163
29	Heinrich Klaasen	150
66	Rajat Patidar	149
105	Virat Kohli	142
21	Devdutt Padikkal	136
3	Ajinkya Rahane	131

In [31]:

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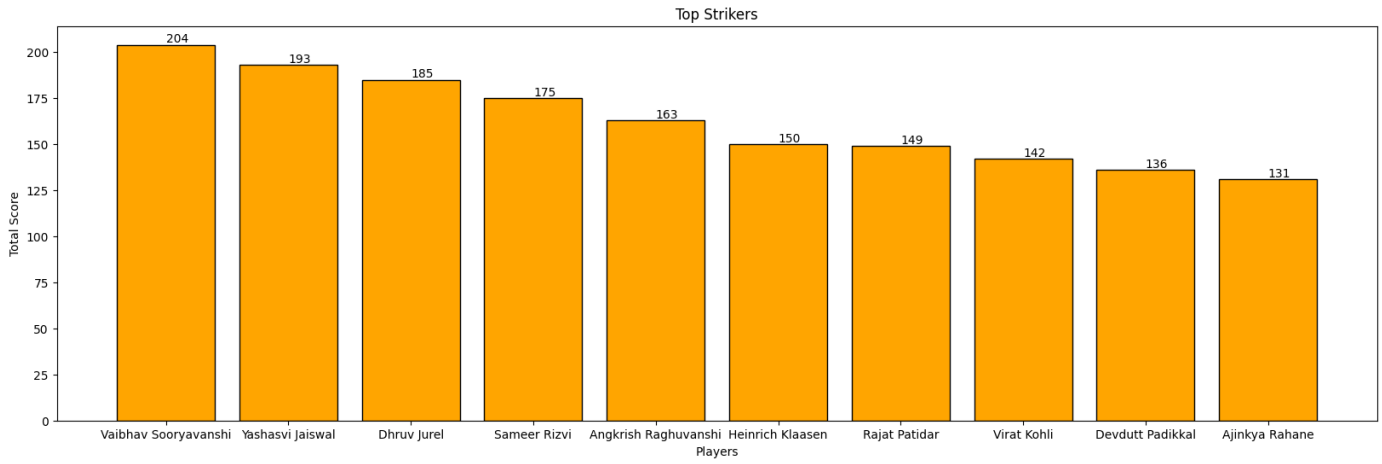
```
plt.figure(figsize=(20,6))
plt.bar(df1['striker'], df1['total' ], color='orange', edgecolor='black')
plt.title('Top Strikers')
plt.xlabel('Players')
plt.ylabel('Total Score')
for x, y in zip(df1['striker' ], df1['total']):
    plt.text(x=x,y=y, s=y,va='bottom')
plt.show()
```



Top Batsman boundaries

In [32]:

```
df1 = df.groupby('striker').agg(
total = ('is_six','sum')
).reset_index ()
df1 = df1.sort_values('total',ascending=False).head(10)
df1
```

Out[32]:

	striker	total
101	Vaibhav Sooryavanshi	18
66	Rajat Patidar	13
80	Sameer Rizvi	11
108	Yashasvi Jaiswal	10
77	Ryan Rickelton	9
96	Tim David	9
22	Dhruv Jurel	8
52	Mukul Choudhary	7
73	Rohit Sharma	7
36	Jos Buttler	7

In [33]:

```
plt.figure(figsize=(20,6))
plt.bar(df1['striker'], df1['total'],color='pink', edgecolor='black')
plt.title('Top Strikers')
plt.xlabel('Players')
```

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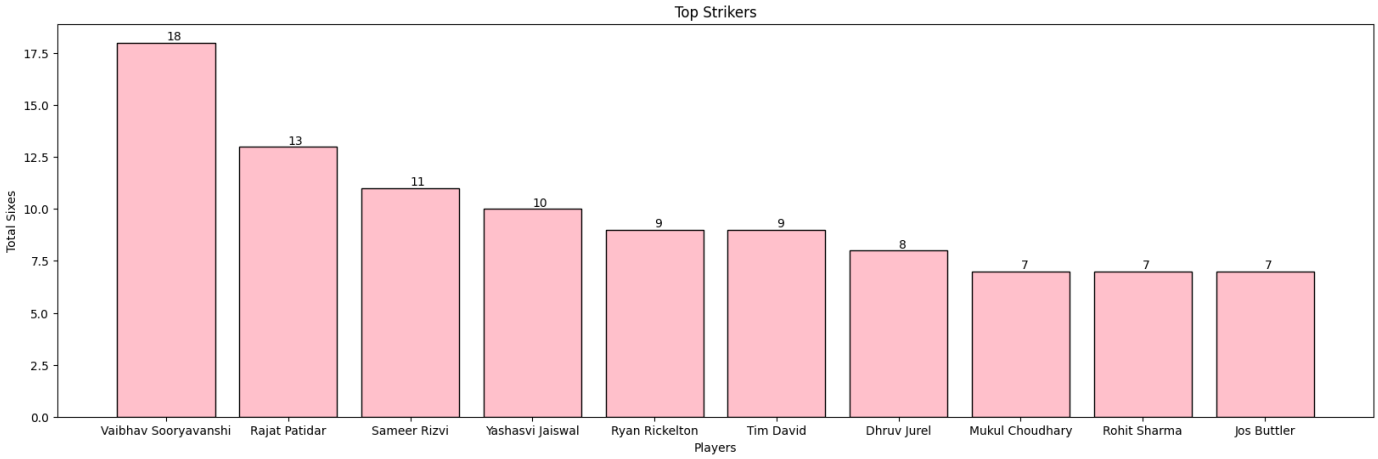
```
plt.ylabel('Total Sixes')

for x,y in zip(df1['striker'],df1['total']):
    plt.text(x=x,y=y,s=y,va='bottom')
plt.show()
```



Strike rate for the top batsman

1. formulae : (runs score / ball faced) * 100
2. wide ball != balls faced

In [34]:

```
# Filter only legal deliveries
legal_balls = df[df['extras'] == 0]

# Groupby on legal balls
df1 = legal_balls.groupby('striker').agg(
    runs=('runs_of_bat', 'sum'),
    balls=('runs_of_bat', 'count')
).reset_index()

# Filter players
df1 = df1[df1['balls'] > 30]

# Strike rate
df1['strike_rate'] = (df1['runs'] / df1['balls']) * 100

# Sort and take top 10
df1 = df1.sort_values('strike_rate', ascending=False).head(10)

# Clean values
df1['strike_rate'] = df1['strike_rate'].round(1)

df1
```

Out[34]:

	striker	runs	balls	strike_rate
101	Vaibhav Sooryavanshi	200	74	270.3
96	Tim David	99	43	230.2
21	Devdutt Padikkal	125	58	215.5

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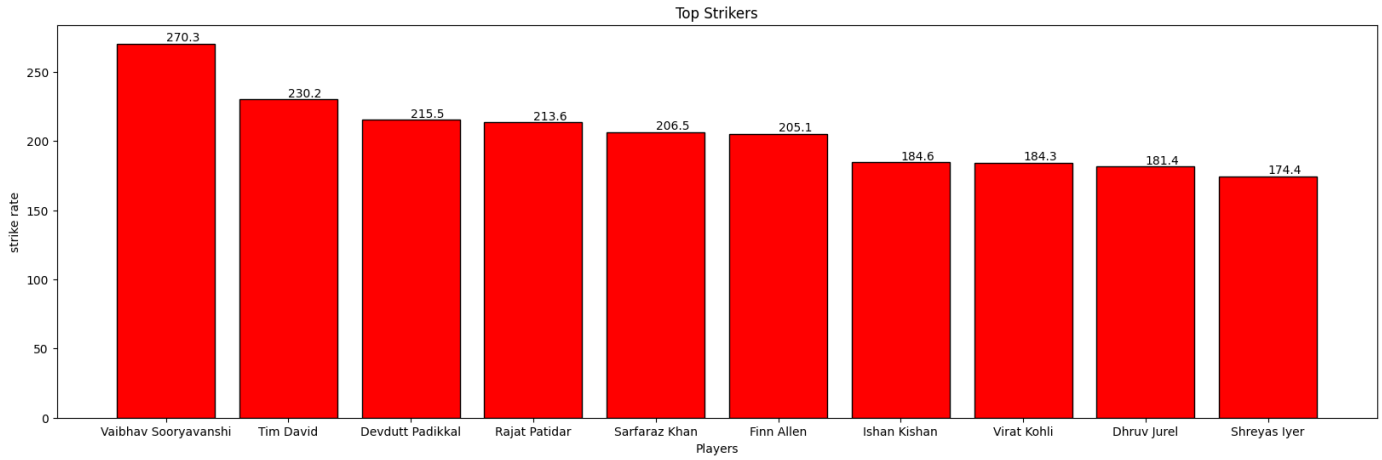
	striker	runs	balls	strike_rate
66	Rajat Patidar	141	66	213.6
82	Sarfaraz Khan	95	46	206.5
24	Finn Allen	80	39	205.1
30	Ishan Kishan	96	52	184.6
105	Virat Kohli	129	70	184.3
22	Dhruv Jurel	176	97	181.4
91	Shreyas Iyer	68	39	174.4

In [35]:

```
plt.figure(figsize=(20,6))
plt.bar(df1['striker'], df1['strike_rate'], color='red', edgecolor='black')
plt.title('Top Strikers')
plt.xlabel('Players')
plt.ylabel('strike rate')

for x,y in zip(df1['striker'], df1['strike_rate']):
    plt.text(x=x, y=y, s=y, va='bottom')

plt.show()
```



Bowler Analysis

Most wickets in IPL 2026 - Bowlers

run-outs are not credited to the bowler

In [36]:

```
df
```

Out[36]:

	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
0	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
1	202601	2026	Group Stage	1	Mar 28,	M.Chinnaswamy Stadium,	SRH	RCB	1

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	match_id	season	phase	match_no	date	venue	batting_team	bowling_team	innings
					2026	Bengaluru			
2	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
3	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
4	202601	2026	Group Stage	1	Mar 28, 2026	M.Chinnaswamy Stadium, Bengaluru	SRH	RCB	1
...	...	...	...	...	...	...	...	...	...
3512	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3513	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3514	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3515	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2
3516	202616	2026	Group Stage	16	Apr 10, 2026	Barsapara Cricket Stadium, Guwahati	RR	RCB	2

3517 rows × 26 columns

In [37]:

```
bowler_wickets = df.loc[ (df['wicket_type'].notnull()) & (df['wicket_type']!='runout') ]
df1 = bowler_wickets.groupby('bowler').agg(
total = ('is_wicket','sum')
).reset_index()

df1 = df1.sort_values('total',ascending=False).head(10)

df1
```

Out[37]:

	bowler	total
43	Ravi Bishnoi	9
19	Jacob Duffy	6
40	Prasidh Krishna	6
2	Anshul Kamboj	5
54	Vaibhav Arora	5
42	Rashid Khan	5
41	Prince Yadav	5

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	bowler	total
37	Nandre Burger	5
36	Mukesh Kumar	5
28	Lungi Ngidi	5

In [38]:

```
df['wicket_type' ]. unique()
```

Out[38]:

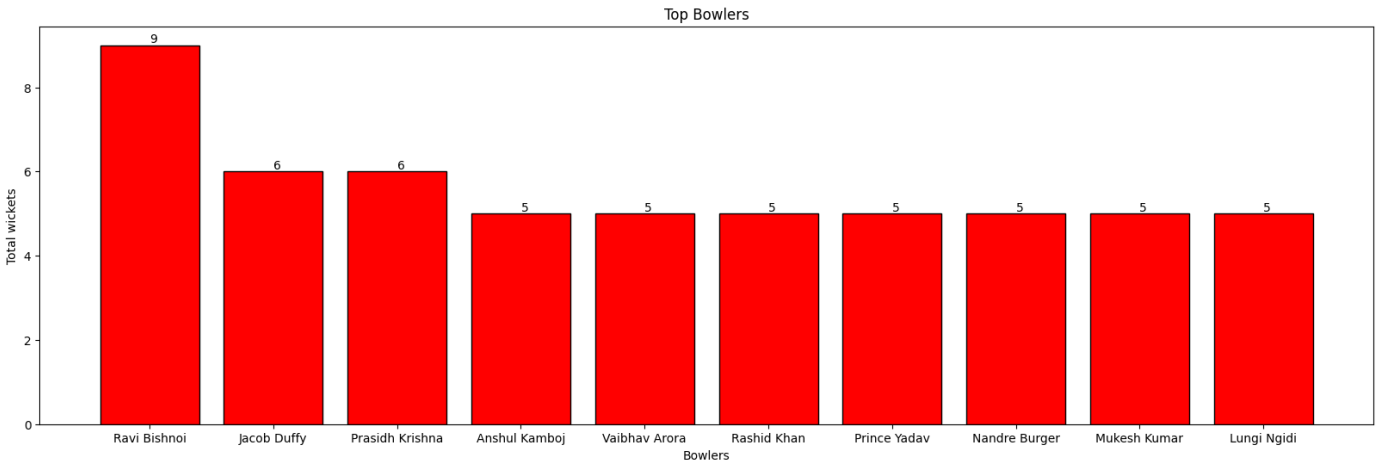
```
<StringArray>
[nan, 'caught', 'bowled', 'runout', 'lbw', 'stumped']
Length: 6, dtype: str
```

In [39]:

```
plt.figure(figsize=(20,6))
plt.bar(df1['bowler'], df1['total'],color='red', edgecolor='black')
plt.title('Top Bowlers')
plt.xlabel('Bowlers')
plt.ylabel('Total wickets')

for x,y in zip(df1['bowler'],df1['total']):
    plt.text(x=x,y=y, s=y,va='bottom' )

plt.show()
```



Which Bowler has best economy gate(runs given per over)

In [40]:

```
legal_balls = df.loc[ (df['wide']==0) & (df['noballs']==0) ]

df1 = df.groupby('bowler').agg(
total = ('runs_of_bat','sum'),
balls = ('runs_of_bat', 'count')
).reset_index()
df1 = df1. loc[ df1['balls']>18 ]
df1['economy_rate'] = (df1['total']/df1['balls'])*6
df1 = df1.sort_values('economy_rate').head(10)
df1
```

Out[40]:

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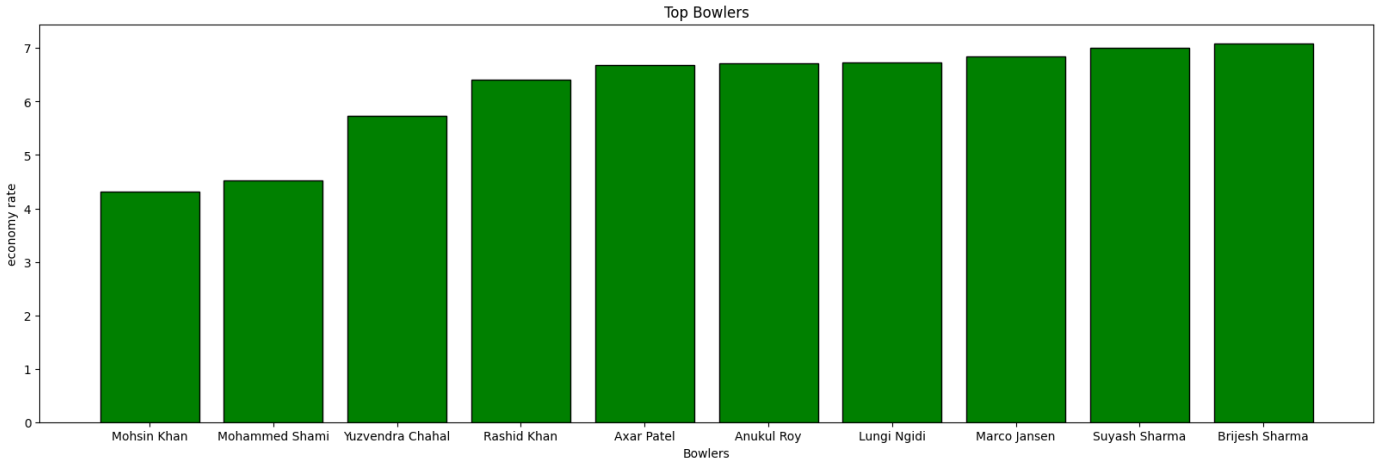
	bowler	total	balls	economy_rate
45	Mohsin Khan	18	25	4.320000
43	Mohammed Shami	58	77	4.519481
76	Yuzvendra Chahal	44	46	5.739130
55	Rashid Khan	79	74	6.405405
11	Axar Patel	69	62	6.677419
7	Anukul Roy	56	50	6.720000
35	Lungi Ngidi	82	73	6.739726
37	Marco Jansen	57	50	6.840000
66	Suyash Sharma	49	42	7.000000
14	Brijesh Sharma	52	44	7.090909

In [41]:

```
plt.figure(figsize=(20,6))
plt.bar(df1['bowler'],df1['economy_rate'], color='green', edgecolor='black')
plt.title('Top Bowlers')
plt.xlabel('Bowlers')
plt.ylabel('economy rate')
plt.show()
```



How are batsman getting out(wicket type)

In [42]:

```
df1 = df['wicket_type'].value_counts().reset_index()
df1
```

Out[42]:

	wicket_type	count
0	caught	141
1	bowled	29
2	runout	13
3	lbw	7
4	stumped	2

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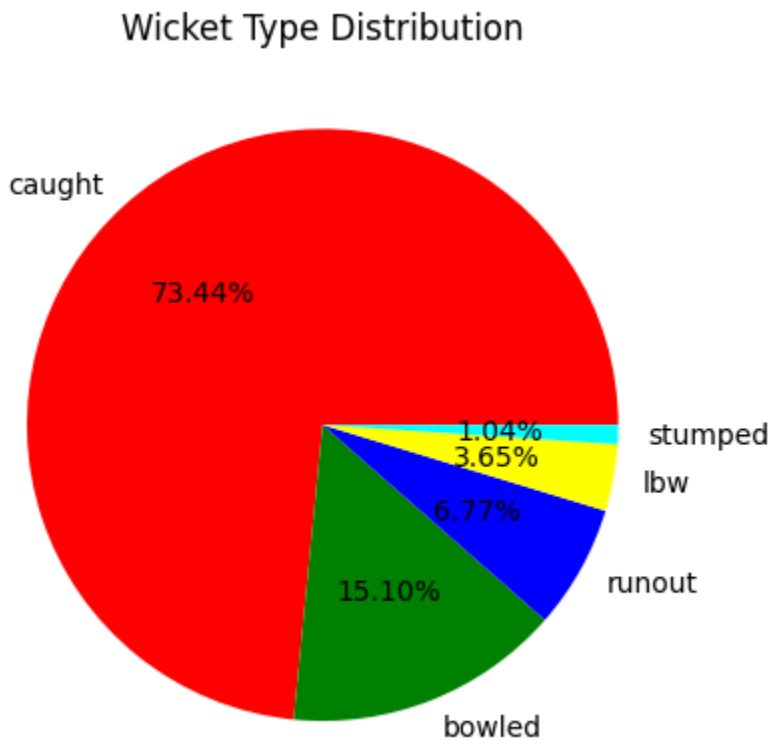
In [43]:

```
plt.pie(
    df1['count'],
    labels=df1['wicket_type'],
    autopct='%0.2f%%',
    colors=['red', 'green', 'blue', 'yellow', 'cyan']
)

plt.title("Wicket Type Distribution")
plt.show()
```



Average Run rate over by over

In [44]:

```
# Create over number
df['over_number'] = df['over'].astype(int)

# Group by overs
edf = df.groupby('over_number').agg(
    avg_runs=('Total_runs', 'mean')
).reset_index()

# Round values (apply to edf, not df1)
edf['avg_runs'] = edf['avg_runs'].round(2)

edf
```

Out[44]:

	over_number	avg_runs
0	1	1.48
1	2	1.46

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	over_number	avg_runs
2	3	1.57
3	4	1.65
4	5	1.78
5	6	1.68
6	7	1.48
7	8	1.49
8	9	1.62
9	10	1.25
10	11	1.45
11	12	1.71
12	13	1.42
13	14	1.44
14	15	1.69
15	16	1.56
16	17	1.65
17	18	1.53
18	19	1.79
19	20	1.81

In [45]:

```
plt.figure(figsize=(20,6))

plt.plot(edf['over_number'], edf['avg_runs'],
         marker='o', linewidth=2)

plt.title('Average Runs per Over')
plt.xlabel('Over')
plt.ylabel('Average Runs')

plt.xticks(np.arange(1, 21))

# Add labels
for x, y in zip(edf['over_number'], edf['avg_runs']):
    plt.text(x=x, y=y, s=round(y,2), va='bottom', ha='center')

plt.grid()
plt.show()
```

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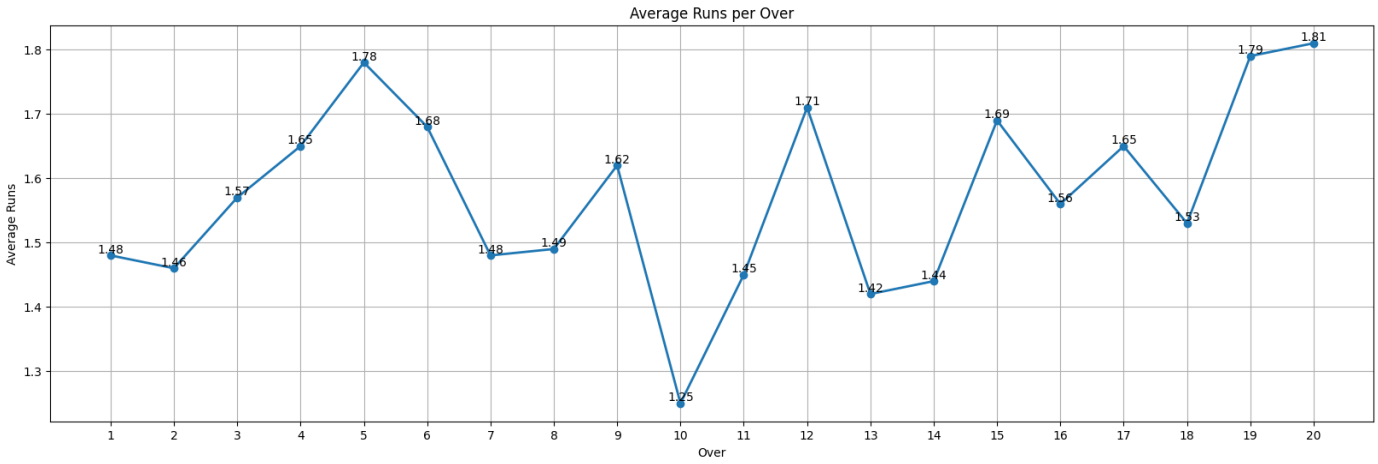

Which team is better in Powerplay , middle or in death overs ?

In [46]:

```
df1 = df.groupby(['batting_team','phase_group']).agg(
total = ('Total_runs', 'sum' )
).reset_index()

df1
```

Out[46]:

	batting_team	phase_group	total
0	CSK	Death(16-20)	171
1	CSK	Middle(7-15)	224
2	CSK	Powerplay(1-6)	149
3	DC	Death(16-20)	156
4	DC	Middle(7-15)	252
5	DC	Powerplay(1-6)	108
6	GT	Death(16-20)	160
7	GT	Middle(7-15)	261
8	GT	Powerplay(1-6)	155
9	KKR	Death(16-20)	148
10	KKR	Middle(7-15)	238
11	KKR	Powerplay(1-6)	200
12	LSG	Death(16-20)	148
13	LSG	Middle(7-15)	211
14	LSG	Powerplay(1-6)	124
15	MI	Death(16-20)	111
16	MI	Middle(7-15)	256
17	MI	Powerplay(1-6)	142
18	PBKS	Death(16-20)	109

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	batting_team	phase_group	total
19	PBKS	Middle(7-15)	153
20	PBKS	Powerplay(1-6)	112
21	RCB	Death(16-20)	208
22	RCB	Middle(7-15)	278
23	RCB	Powerplay(1-6)	168
24	RR	Death(16-20)	101
25	RR	Middle(7-15)	324
26	RR	Powerplay(1-6)	265
27	SRH	Death(16-20)	201
28	SRH	Middle(7-15)	264
29	SRH	Powerplay(1-6)	120

In [47]:

```
df1 = df.pivot_table(index='batting_team' , columns='phase_group', values='Total_runs' ,
df1
```

Out[47]:

phase_group	batting_team	Death(16-20)	Middle(7-15)	Powerplay(1-6)
0	CSK	171	224	149
1	DC	156	252	108
2	GT	160	261	155
3	KKR	148	238	200
4	LSG	148	211	124
5	MI	111	256	142
6	PBKS	109	153	112
7	RCB	208	278	168
8	RR	101	324	265
9	SRH	201	264	120

In [49]:

```
plt.figure(figsize=(15,6))

xval = np.arange(1,11) #[1,2,3,4,5, 6, 7, 8, 9,10]

plt.bar(xval, df1['Powerplay(1-6)'],width=0.3,color='green',label='Powerplay(1-6)',edgec
#[1 2 3 4 5 6 7 8 9 10]
plt.bar(xval+0.2,df1['Middle(7-15)'],width=0.3,color='lightgreen',label='Middle(7-15)',
edgecolor='black') #[1.2,2.2,3.2,4.2,5.2,6.2,7.2,8.2,9.2,10.2]
plt.bar(xval+0.4,df1['Death(16-20)'],width=0.3,color='blue',label='Reath(16-20)',edgecol
plt. legend()

for x, y in zip(xval, df1['Powerplay(1-6)']):
```

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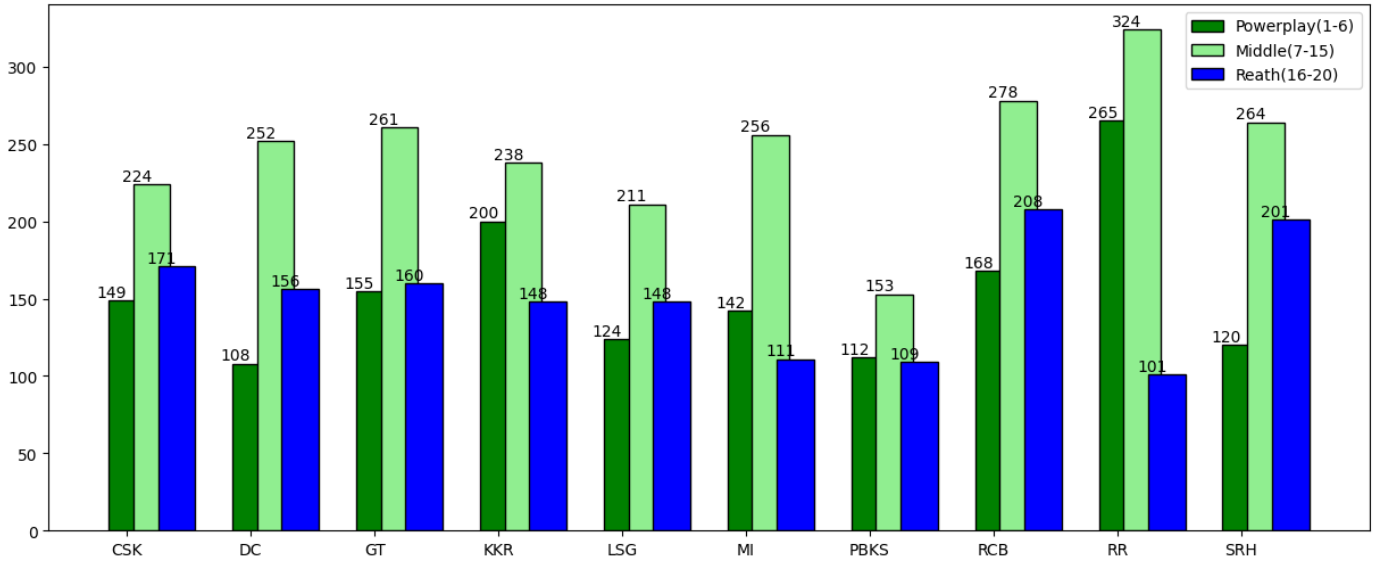
```
    plt. text (x=x,y=y, s=y, va='bottom',ha='right')
for x,y in zip(xval+0.2,df1['Middle(7-15)']):
    plt.text(x=x,y=y,s=y, va='bottom',ha='right')

for x,y in zip(xval+0.4,df1['Death(16-20)']):
    plt.text(x=x,y=y,s=y,va='bottom',ha='right')

plt.xticks(xval, df1['batting_team'])

plt.show()
```



In [ ]:

In [ ]:

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